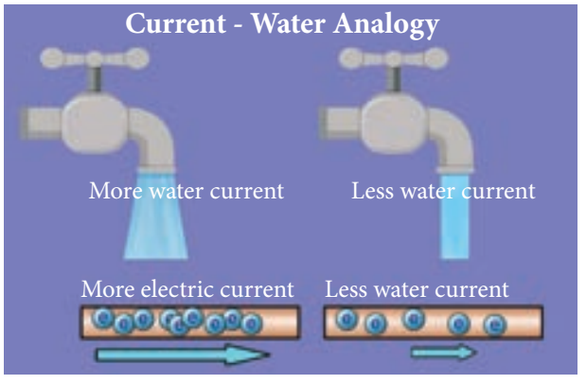
Let us see the analogy of flow of electric current with the water flow.

Water flowing through pipes is pretty good mechanical system that is a lot like an electrical circuit. This mechanical system consists of a pump pushing water through a closed pipe. Imagine that the electrical current is similar to the water flowing through the pipe. The following parts of the two systems are related

- The pipe is like the wire in the electric circuit and the pump is like the battery.
- The pressure generated by the pump drives water through the pipe.
- The pressure is like the voltage generated by the battery which drives electrons through the electric circuit.
- Suppose, there are some dust and rust that plug up the pipe and slow the flow of water, creating a pressure difference from one end to the other end of the pipe. In similar way, the resistance in the electric circuit resists the flow of electrons and creates a voltage drop from one end to the other. Energy loss is shown in the form of heat across the resistor.



2.3. Sources of Electric current - Electro chemical cells or electric cells

An electric cell is something that provides electricity to different devices that are not fed directly or easily by the supply of electricity.

ACTIVITY 2

Shall we produce electricity at our home?

Materials required:

Zinc and copper electrodes, a light blub, connecting wires, and fruits such as lemons, orange, apples, grapes, and bananas.

Procedure:

1. Set up a circuit as shown in figure
2. Note the brightness of the blub when the circuit is connected to a lemon.
3. Repeat the experiment using the other fruits listed above. Do you notice the differences in the brightness of the bulb when it is connected to different fruits? Which fruit gives the greatest brightness? Why? (If you do not know please get the appropriate reason from your teacher)

Inference:

In the above activity what makes enabled the bulb to glow. Why there is a difference in the brightness of the bulb? The reason is that the fruits which you have connected to the bulb produces the electric energy at different levels

The sources which produce the small amount of electricity for shorter periods of time is called as electric cell or electro chemical cells. Electric cell converts chemical energy into electrical energy

In addition to electro chemical, we use electro thermal source for generating electricity for large scale use.

It has two terminals. When electric cells are used, a chemical reaction takes place inside the cells which produces charge in the cell.

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